

ASSIGNMENT-3.5

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

Answer: -

1. Test Communication between Laptop and Phone

1. Connect **both the laptop and phone to the same Wi-Fi network.**
2. On the phone, find its local IP address:
 - o **Android:** Wi-Fi → connected network → Network details → **IP address.**
 - o Example: 192.168.1.105
3. On the laptop, open **Command Prompt.**
4. Run:

```
Ping 192.168.1.105
```

Replace the example IP with your phone's actual IP address.

5. Check the result:
 - o **Reply from...** → the laptop and phone can communicate.
 - o **Request timed out** → communication may be blocked by the phone, router, or Wi-Fi isolation settings.

Assignment Result

Result: Both devices were connected to the same Wi-Fi network. The laptop successfully pinged the phone's IP address and received replies, confirming that the two devices could communicate over the local network.

If you have a screenshot of your ping result, upload it and I can tell you exactly what result to write.

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Answer: -

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:482c:400::]
over a maximum of 30 hops:

  1    7 ms    3 ms    2 ms  2409:40c1:643e:2da8::28
  2    *      *      *      Request timed out.
  3   22 ms   28 ms   20 ms  2405:200:5210:e:3925::ff03
  4   22 ms   16 ms   18 ms  2405:200:81a:1513:61::4
  5    *      *      *      Request timed out.
  6   35 ms   17 ms   19 ms  2405:200:801:200::881c
  7    *      *      *      Request timed out.
  8   38 ms   27 ms   28 ms  2001:4860:1:1::1b8e
  9   39 ms   26 ms   31 ms  2001:4860:1:1::1b8e
 10   35 ms   31 ms   24 ms  2001:4860:482c:400::

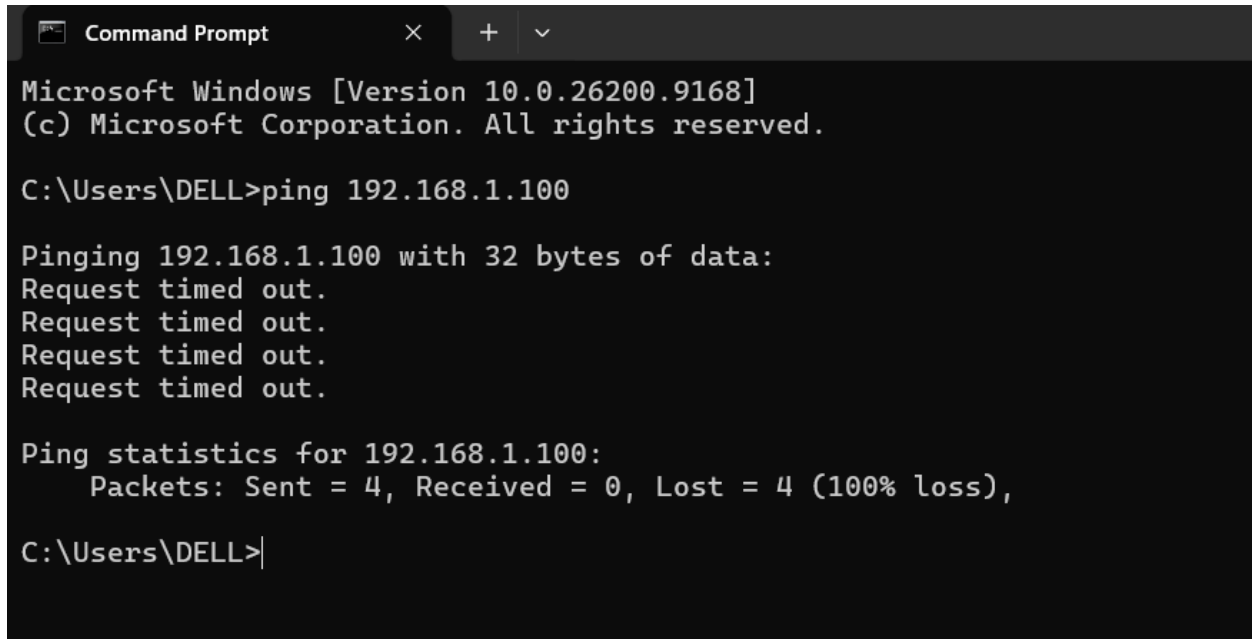
Trace complete.

C:\Users\DELL>
```

Number Of Hops :- 10.

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

Answer: -

A screenshot of a Windows Command Prompt window. The title bar shows 'Command Prompt' with standard window controls. The text inside the window shows the Windows version and copyright information, followed by a command to ping 192.168.1.100. The output shows four 'Request timed out' messages and a summary indicating a 100% loss of packets.

```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>ping 192.168.1.100

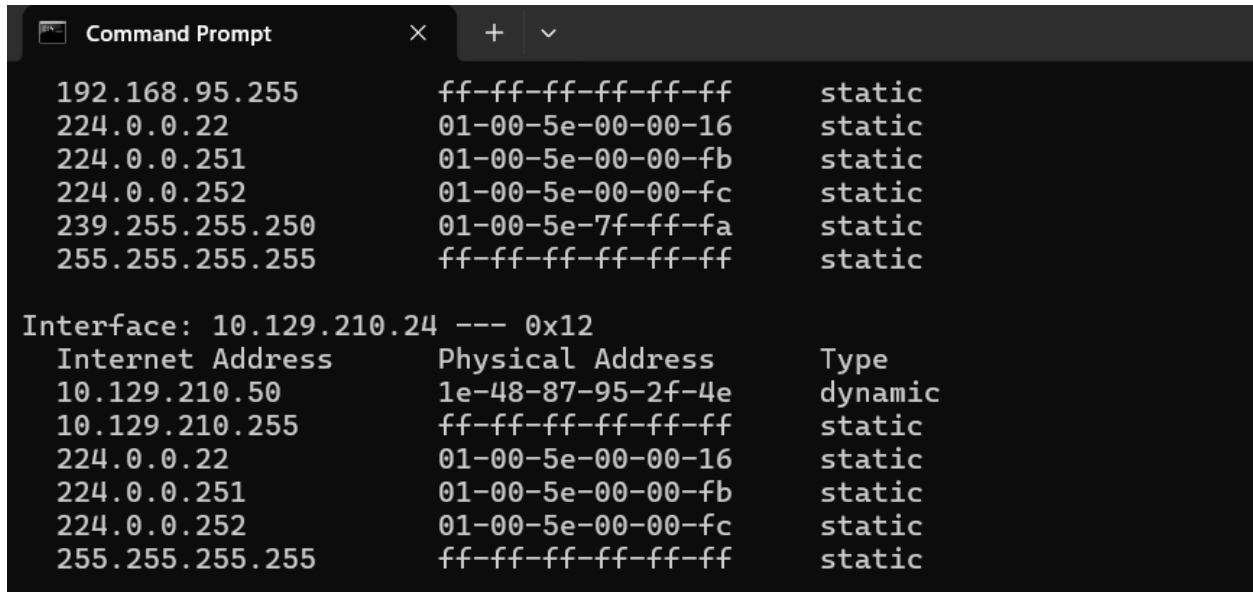
Pinging 192.168.1.100 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\DELL>
```

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Answer: -



```
Command Prompt
192.168.95.255      ff-ff-ff-ff-ff-ff      static
224.0.0.22         01-00-5e-00-00-16      static
224.0.0.251        01-00-5e-00-00-fb      static
224.0.0.252        01-00-5e-00-00-fc      static
239.255.255.250    01-00-5e-7f-ff-fa      static
255.255.255.255    ff-ff-ff-ff-ff-ff      static

Interface: 10.129.210.24 --- 0x12
Internet Address   Physical Address       Type
10.129.210.50      1e-48-87-95-2f-4e      dynamic
10.129.210.255     ff-ff-ff-ff-ff-ff      static
224.0.0.22         01-00-5e-00-00-16      static
224.0.0.251        01-00-5e-00-00-fb      static
224.0.0.252        01-00-5e-00-00-fc      static
255.255.255.255    ff-ff-ff-ff-ff-ff      static
```

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.

Answer: -

Network Configuration :

The network was configured using a wireless connection. A laptop was connected to the network through a Wi-Fi adapter. The system automatically obtained an IP address from the router using DHCP. After connecting to the network, the assigned IP address and network details were verified using standard Windows networking commands.

```
C:\Users\Admin>arp -a

Interface: 192.168.30.1 --- 0x7
    Internet Address      Physical Address      Type
    192.168.30.254        00-50-56-ff-9b-b2    dynamic
    192.168.30.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    231.237.113.100       01-00-5e-6d-71-64    static
    239.192.152.143       01-00-5e-40-98-8f    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.133.1 --- 0x10
    Internet Address      Physical Address      Type
    192.168.133.254       00-50-56-e9-17-a6    dynamic
    192.168.133.255       ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    231.237.113.100       01-00-5e-6d-71-64    static
    239.192.152.143       01-00-5e-40-98-8f    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

Connectivity Testing Using Ping :

```
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Request timed out.
Request timed out.
```

Here is the complete text output formatted to match your image:

Observation

The ping test failed and displayed "**Request timed out**" messages. This indicated that the destination device or network path was unavailable. The packet loss rate was 100%, confirming that communication could not be established.

Route Testing Using Traceroute:

```
C:\Users\Admin>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1    2 ms    2 ms    2 ms  10.41.171.31
  2   189 ms   202 ms   202 ms  192.168.17.10
  3    52 ms    41 ms    30 ms  192.168.16.33
  4    34 ms    21 ms    34 ms  192.168.19.3
  5    26 ms    32 ms    33 ms  192.168.19.33
  6    37 ms    48 ms    27 ms  192.168.252.42
  7    26 ms    17 ms    30 ms  125.20.216.10
  8    42 ms   100 ms    40 ms  125.20.216.9
  9    36 ms    40 ms    49 ms  116.119.73.209
 10    43 ms    73 ms    46 ms  72.14.212.48
 11    52 ms    69 ms    43 ms  142.251.225.67
 12    52 ms    35 ms    39 ms  192.178.86.247
 13   96 ms    41 ms    38 ms  dns.google [8.8.8.8]

Trace complete.
```

Issues:

1. Request Timed Out Errors

- Occurred when a device was disconnected from the network.
- Resulted in 100% packet loss during ping testing.

2. Network Reachability Problems

- Communication failed when the target device was offline.
- Connectivity was restored after reconnecting the device.

3. Potential Causes

- Device disconnected from Wi-Fi.
- Firewall blocking ICMP requests.
- Temporary network interruption.